MARKING SCHEME
FORM FOUR PHYSICS 01

01.

I	II	III	IV	V	VI	VII	VIII	IX	X
D	B	D	D	C	B	C	B	C	C

1@ = 10 marks

02.

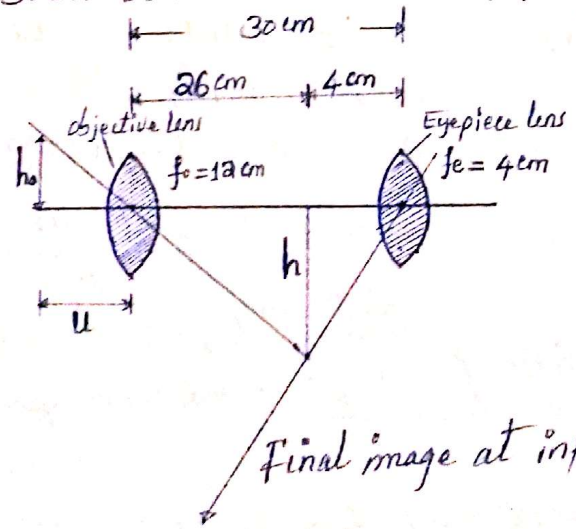
I	II	III	IV	V	VI
H	A	G	B	D	F

1@ = 6 marks.

03. (a) Explanation: The direction of the ray of light changes when it enters and exists the glass block due to refraction. This is simply because of difference in medium, when the ray of light is from the air of less dense and when entering in the glass block of higher dense ..... (4 marks)

(b). Solution.

The final image is at infinity and two lenses are separated by 30cm distance as shown in figure.



..... (1 mark)

Now: from

$V = L - f_e$ ..... ($\frac{001}{2}$ mark)

$V = 30cm - 4cm = 26cm$ ..... ($\frac{001}{2}$ mark)

Using lens formula.

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$; $\frac{1}{12} = \frac{1}{u} + \frac{1}{26}$ ..... (01 mark)

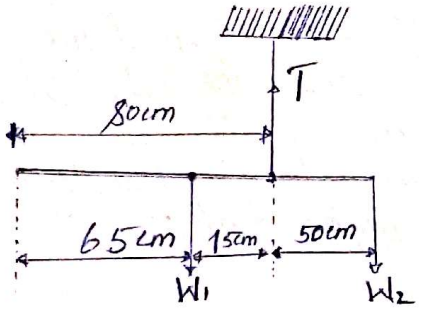
$\frac{1}{u} = \frac{1}{12cm} - \frac{1}{26cm}$; $\frac{1}{u} = \frac{26cm - 12cm}{312cm^2}$ ..... (01 mark)

$\frac{1}{u} = \frac{14}{312cm}$; $u = 22.3\,cm$ ..... (01 mark)

∴ The object distance is 22.3 cm from the objective lens.

04. (a). Is not be practical to make a barometer with water this is because (i). Water wets glass ..... (01mark).
(ii). Water is colourless. ..... (01mark).
(iii). Water is a poor conductor of heat (1mark)
(iv). It boils at 100°C ..... (01mark).

(b). Solution.
Consider the diagram below.



from the principle of moment ..... (01mark)/2
Sum of clock wise moment = Sum of anticlock wise moment.
$W_2d_2 = W_1d_1$ or
$m_2d_2 = m_1d_1$
$m_2 \times 50cm = 120g \times 15cm$ ---- (01mark)/2
$m_2 = \frac{120g \times 15cm}{50cm}$ ..... (01mark)/2
$m_2 = 36g$ or $0.036kg$ ---- (01mark)/2
Therefore a mass of 36g required for a system to be in equilibrium.
..... (001mark)/2

Again: Weight = mass x acceleration due to gravity
$W_1 = mg$
$= 0.12kg \times 10N/kg$
$W_1 = 1.2N$ ..... (001/2 mark)
$W_2 = m_2 \times g$
$= 0.036kg \times 10N/kg$
$W_2 = 0.36N$ ..... (001/2 mark)

Tension(T): upward force = Downward force.
$T = W_1 + W_2$ ..... (001/2 mark)
$= 1.2N + 0.36N$ ..... (001/2 mark)
$T = 1.56N$
∴ Tension (T) = 1.56N ..... (001/2 mark)

05. (a) Explanation: Here the factors required are conditions of floating and sinking. ..... (01 mark)

Conditions for floating.

i. The volume of the leaf must be large enough to displace a large volume of water. ..... ($\frac{01}{2}$ mark)

ii. The density of leaf must be less than the density of water. ..... ($\frac{01}{2}$ mark)

iii. The upward force (upthrust) must be greater or equal to the total weight of leaf. ..... ($\frac{01}{2}$ mark)

Condition for sinking.

i. Volume of stone must be small ..... ($\frac{01}{2}$ mark)

ii. The density of stone must be greater than density of water. ..... ($\frac{01}{2}$ mark)

iii. Weight of stone must be greater than the up thrust. ..... ($\frac{01}{2}$ mark)

(b). SOLUTION

Given that:

Input force (F_1) = 11 N

distance moved by effort (d_1) = 0.4 m.

output force / Load (F_2) = 40N

distance moved by Load (d_2) = 0.1 m.

Efficiency of machine (e) = ?

from

$$\text{Efficiency } (e) = \frac{\text{work output}}{\text{work input}} \times 100\% \quad \text{..... (01 mark)}$$

- work output = Load x distance moved by load.

$$= 40N \times 0.1m$$

$$(W_o) = 4J. \quad \text{..... (01 mark)}$$

- Work input (W_i) = Effort x distance moved by effort

$$= 11N \times 0.4m$$

$$= 4.4J. \quad \text{..... (01 mark)}$$

05. (b) efficiency (e) = $\frac{\text{work output (wo)}}{\text{work input wi}} \times 100\%$

$$e = \frac{4\text{J}}{4.4\text{J}} \times 100\% \quad \text{---------- (01 mark)}$$

$$e = 90.9\%$$

∴ The efficiency of machine is 90.9% ---- (01 mark)

06. (a) Explanation:

(i). The bimetallic strip acts as a temperature - sensitive switch, opening and closing the circuit based on temperature changes..... (2marks)

(ii). The circuit controls temperature by heating the room when the switch is closed and the bimetallic strip automatically turns the heater off and on to maintain a set temperature..... (2marks)

(b). Smoke doesn't always rise due to air pressure, differences, wind, temperature inversions and varying smoke density..... (05marks)

07. (a) Explanation: Sound waves need a medium (like air) to travel. Space is a vacuum, lacking this medium. Therefore, sounds from other planets can't reach Earth. ..... (4marks)

(b). Solution.

Different isotopes will differ at least by 1 neutron. That is,

$A_1 = 108$, $A_2 = 109$, $A_3 = 110$ ..... A_n

Since there are 25 isotopes, $n = 25$

from arithmetic progression,

$A_n = A_1 + (n-1)d$ ---------- (01mark)

In this case, $A_1 = 108$, $n = 25$, $d = 1$

$\therefore A_n = 108 + (25-1)1$ ---------- (01mark)

$= 132$ ---------- (01mark)

Hence, the heaviest isotope by the symbol of tin will be $^{132}_{50}Sn$ ---------- (02marks)

08. (a) Answer: P-waves were recorded first. P-waves travel faster than S-waves through the earth's interior, meaning they will cover the distance between Mbeya and Dodoma more quickly. ---------- (3marks).

(b). Answer: Three uses of constellations are.

i. Navigation: Constellations provide celestial landmarks for determining direction and location, particularly useful for seafaring and exploration ---------- (2marks)

ii. Time keeping: The seasonal changes in the visibility of constellations allow for tracking the passage of time and predicting seasonal changes. ---------- (2marks)

08. (b) iii/. Cultural and mythological significance:
Constellations are integral to many cultures' mythologies and storytelling, offering insights into their history and beliefs ..... (2marks)

SECTION C: 30marks.

09. (a) Answer: As the hand approaches and moves across the charged plate of the electroscope, the leaves will partially collapse due to the induced charge separation and partial neutralization of the electroscope's charge: ..... (05marks).

(b). Solution:
Given that.
Emf of the first cell = 6V
Current(I) in a series = 3A
Emf of the second cell(E_2) = 1.5V
Internal resistance(r_2) = Neg.
from
$E = E_1 + E_2$ in series.
For first cell.
$E_1 = I(R + r_1)$ ..... ($\frac{1}{2}$ mark)
$r_1 = 0$
$E_1 = V = IR$ ..... ($\frac{1}{2}$ mark)
$6V = 3R$.
$\frac{6V}{3} = R$ ..... ($\frac{1}{2}$ mark)
$2\Omega = R$ ..... ($\frac{1}{2}$ mark)
When the second cell is added then
total emf (E) = $E_1 + E_2$ ..... ($\frac{1}{2}$ mark)
$E = 6V + 1.5V$
$E = 7.5V$ ..... ($\frac{1}{2}$ mark)
But $E = I(R + r_2)$

Q9. (b). $E = I(R + r_2)$ ---------- ($\frac{001}{2}$ mark)

$7.5V = 3A(R + r_2)$ ---------- ($\frac{001}{2}$ mark)

$\frac{7.5V}{3A} = 2 + r_2$

$2.5\Omega = 2\Omega + r_2$

$2.5\Omega - 2\Omega = r_2$

$0.5\Omega = r_2$ ---------- ($\frac{001}{2}$ mark)

∴ The internal resistance of the second cell is 0.5Ω ---------- ($\frac{001}{2}$ mark)

(C). Solution:

Given that:

Supplied voltage(V) = 240V.

power of motor(P_m) = 250W (0.25kW).

power of heating element(P_H) = 3kW.

Time (t) = 1.5 hour per week.

Annual cost = Required.

Motor:

Energy consumption per week(E_m)

$E = Pt$.

$E_m = P_m t$ ---------- ($\frac{001}{2}$ mark)

$E_m = 0.25kW \times 1.5h$

$= 0.375 kWh$. ---------- ($\frac{001}{2}$ mark)

Motor energy consumption per week is 0.375kWh.

Heating element:

Energy consumption per week(E_H)

09. (c). $E_H = P_H \times t$ ---------- ($\frac{001}{2}$ mark)

$E_H = 3kW \times \frac{2}{3} \times 1.5 hour$.

$E_H = 3kW \times 1 hour$.

$E_H = 3kWh$ ---------- ($\frac{001}{2}$ mark)

∴ Heating element energy consumption per week is 3kWh.

Total weekly energy consumption $= E_m + E_H$

$= 0.375kWh + 3kWh$.

$= 3.375kWh$ ---------- (01 mark)

Energy consumption per week $= 3.375kWh$.

Annual energy consumption.

1 week $= 3.375 kWh$.

52 week $= X$

$$X = \frac{52 week \times 3.375 kWh}{1 week}$$

$X = 175.5 kWh$ per year. (01 mark)

Annual cost:

1kWh $= 5 Tsh$.

175.5kWh $= X$

$$X = \frac{175.5 kWh \times 5 Tsh}{1 kWh}$$

$X = 877.5 Tsh$.

∴ The annual cost is 877.5 Tsh ---- (01 mark)

10 (a) Answer: A CRT uses thermionic emission from a heated cathode to produce a beam of electrons accelerated towards a positive anode in a vacuum. The beam is deflected by electric or magnetic fields to scan fluorescent screen, producing an image. ---------- (2marks)

i. A hotter filament increases beam intensity. (01 mark)

ii. Increasing anode voltage increases beam speed and energy. ---------- (01 mark)

(b). Solution:

Given that:

Half life($t_{\frac{1}{2}}$) = 10 minute

initial radioactive(N_0) = 100%

final radioactive (N) = 10%

Time taken (T) = ?eq

from

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^n \text{ ---------- } \left(\frac{001}{2} \text{ mark}\right)$$

$$\frac{10\%}{100\%} = \left(\frac{1}{2}\right)^n \text{ ---------- } \left(\frac{001}{2} \text{ mark}\right)$$

$$\frac{1}{10} = \frac{1}{2^n}$$

$$2^n = 10 \text{ ---------- } \left(\frac{001}{2} \text{ mark}\right)$$

Apply log both side.

$$\log 2^n = \log 10 \text{ ---------- } \left(\frac{001}{2} \text{ mark}\right)$$

$$n \log 2 = \log 10$$

$$n = \frac{\log 10}{\log 2}$$

$$n = 3.322 \text{ ---------- } \left(\frac{001}{2} \text{ mark}\right)$$

10 (b). from $T = n \times t_{\frac{1}{2}}$ ---------- (001/2 mark)

$T = 3.322 \times 10$ minutes

$T = 33.22$ minutes ---------- (001/2 mark)

∴ The time taken of radioactive material 33.22 min

---------- (001/2 mark)

(c) i/ Answer: Doping is the process of intentionally introducing impurities into a pure semiconductor material to alter its electrical properties. This controlled addition of impurities modifies the number of charge carriers. ---------- (01 mark)

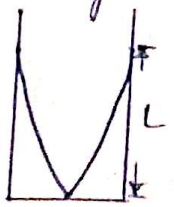
ii/ Answer: Doping a pure semiconductor with pentavalent impurity atom (eg. phosphorus, arsenic, antimony) introduces extra electrons. Four of the impurity atom's valence electrons form covalent bonds with the surrounding semiconductor atom, while the fifth electron becomes a free conduction electron, increasing the material's conductivity and creating an n-type semiconductor. ---------- (04 marks)

iii/ Phosphorus (P) and Arsenic (As) ---------- (02 marks)

11. (a). Closed pipe.

i/ fundamental notes (f_0)

diagram



$L = \frac{\lambda_0}{4}$ ---------- (001/2 mark)

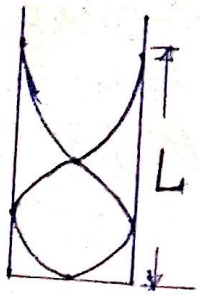
$\lambda_0 = 4L$

from $f_0 = \frac{v}{\lambda_0}$

$f_0 = \frac{v}{4L}$

∴ fundamental notes (f_0) = $\frac{v}{4L}$ ---------- (001/2 mark)

11 (a). first overtone (f_1)



for this case

$L = \frac{3\lambda_1}{4}$, $\lambda_1 = \frac{4L}{3}$ ..... ($\frac{001}{2}$ mark)

from $f_1 = \frac{V}{\lambda_1}$

$f_1 = \frac{3V}{4L}$ ..... ($\frac{001}{2}$ mark)

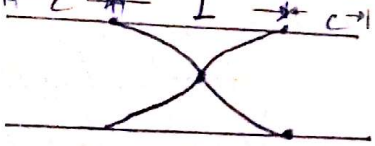
$f_1 = 3\left(\frac{V}{4L}\right)$

$f_1 = 3f_0$.

$\therefore$ first overtone (f_1) $= 3f_0$ ..... ($\frac{001}{2}$ mark)

For oppen pipe.

i/. Consider the diagram.



from the case

$L = \frac{\lambda_0}{2}$ ..... ($\frac{001}{2}$ mark)

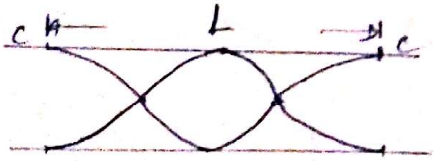
$\lambda_0 = 2L$

$f_0 = \frac{V}{\lambda_0}$

$f_0 = \frac{V}{2L}$

$\therefore$ fundamental notes (f_0) $= \frac{V}{2L}$ ..... ($\frac{001}{2}$ mark)

ii/. first overtone (f_1).



from this case

$L = \frac{2\lambda_1}{2}$

$\lambda_1 = \frac{2L}{2}$.

from $f_1 = \frac{V}{\lambda_1}$

$f_1 = \frac{V}{\lambda_1}$ ..... ($\frac{001}{2}$ mark)

$f_1 = \frac{2V}{2L}$

$f_1 = 2\left(\frac{V}{2L}\right)$ ..... ($\frac{001}{2}$ mark)

$f_1 = 2f_0$ ..... ($\frac{001}{2}$ mark)

$\therefore$ first overtone (f_1) $= 2f_0$.

11 (b) Lenz's law is a special case of the law of conservation of energy because the induced current oppose the change in magnetic flux, requiring energy from the source of the changing magnetic field. ..... (03marks)

(c) (i). A - armature/striker ..... ($\frac{1}{2}$ mark)
B - Gong/bell ..... ($\frac{1}{2}$ mark)
E - Electromagnet ..... ($\frac{1}{2}$ mark)
K - Switch ..... ($\frac{1}{2}$ mark)
T - Contact ..... ($\frac{1}{2}$ mark)
U - Battery ..... ($\frac{1}{2}$ mark)

(ii). When the switch is closed, current flows through the electromagnet, magnetizing it and attracting the armature. The armature strikes the gong and simultaneously breaks the circuit demagnetizing the electromagnet and allowing the armature to spring back, repeating cycle. ..... (04marks)